



Effects and behaviour of experimental fires in grasslands, savannas, and forests of the Brazilian Cerrado

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ABSTRACT

An understanding of fire behaviour and its impacts on natural ecosystems is essential for developing management strategies, specifically for planning prescribed fires. It is well known that vegetation and microclimate factors are important predictors of fire behaviour. However, the relative importance of these factors and how fire behaviour responds to changes in these factors are still understood, especially in the Brazilian Cerrado, which has strong seasonality and high vegetation heterogeneity. Here, we compiled literature data from 65 experimental fire plots in three vegetation types in the Cerrado biome—grasslands, savannas, and forests—corresponding to a gradient of increasing of woody cover. First, we evaluated how aspects of fire behaviour (fire spread rate, fire intensity, and heat released) and of fire-associated emissions (fine fuel consumption, combustion factor, and carbon emissions associated with fine fuel consumption) vary according to vegetation type. Subsequently, we evaluated the relative importance of vegetation (represented by woody cover and fine fuel load) and microclimatic factors (represented by vapour pressure deficit [VPD] and wind speed) in determining the fire behaviour and fire-associated emissions. Using generalised linear models, we evaluated the simultaneous influences of vegetation and microclimate variables (only for independent and non-collinear variables) on fire behaviour and fire-associated emissions. We found that aspects of fire behaviour and fire-associated emissions differed among the vegetation types: compared to forests, grasslands and savannas showed higher fire spread rates and fire intensities, consumed more fine fuel biomass, had larger combustion factors, released more heat, and emitted more carbon. The results show that the vegetation type, VPD and fine fuel load are key metrics for explain fire behaviour and fire-associated emissions in the Cerrado. The VPD was found to be the primary driver of the fire spread rate and fire intensity for all vegetation types and fine fuel load was the primary driver of heat released, fine fuel consumption, combustion factor, and carbon emissions in all vegetation types.

1. Introduction

Fire affects organisms and ecosystems through the release of heat energy during the combustion of plant biomass (Trollope et al., 2002). However, the temporal and spatial variations of heterogeneous landscapes influence the fire behaviour (fire spread rate, fire intensity, and heat released) and thus fire's effects on ecosystems (for example, trace gas emissions) (Lehmann et al., 2014; McKenzie and Kennedy, 2012). Thus, fire management strategies are more effective when it takes the heterogeneity of landscapes into account (Parks et al., 2018; Price et al., 2012).

The advance in knowledge about fire behaviour and the fire-associated emissions have improved the use of fire to manage the landscape. For example, the management of fine fuel has been used to reduce the

risk of wildland fires (unplanned fire) and the extent of burning in South Africa (Van Wilgen et al., 2007; Whitehead et al., 2008) and in Australia (Maraseni et al., 2016). In some areas in South Africa, prescribed fires are carried out at the beginning of the dry season, when the temperature is high ($\geq 30^\circ\text{C}$), has also been used to maintain areas of open vegetation, and promote grazing by large herbivores (Van Wilgen et al., 2007). In Australia, prescribed fires are also carried out at the beginning of the dry season; however, this is when the temperature is lower, which mitigates greenhouse gas emissions (Price et al., 2012). In South America, most fires occur in the Brazilian Cerrado (Fidelis et al., 2018; INPE, 2017). However, the limited information about fire behaviour and effects on Cerrado ecosystems has restricted the adoption of effective fire management strategies (Gomes et al., 2018).

The high environmental heterogeneity of the Cerrado (Sano et al.,

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2019) may influence on different ways the fire behaviour and effects. During the rainy season (October–March) or the transition dry-rainy season (September–October), fires occur naturally through lightning and these fire are less severe than fires in the dry season, because their propagation is normally inhibited by the moisture content of the fuel load and vegetation (Ramos-Neto and Pivello, 2000). During the dry season (April–September), fires occur mainly through human activity and are more severe, reach greater extensions and are more frequent (Pivello, 2011). The type of vegetation also influences the fire behaviour and its effects (Miranda et al., 2010). Grassland and savannas vegetation are more flammable (attributed mainly to higher grass cover and less woody coverage) and fire-tolerant (due to morphological adaptations, such as thick bark of its shrubs and trees) than forest vegetation (Hoffmann et al., 2012). However, fire control and prevention measures are crucial in all environments (grassland, savanna or forest) subject to short fire return intervals (e.g. biennial fires), especially during the dry season, because frequent fires threaten the biomass recovery (Medeiros and Miranda, 2005; Sato, 2003) and increase greenhouse gas emissions (Bustamante et al., 2012). Moreover, in savannas with climatic and edaphic conditions able to support forests, long-term exclusion of fire (~50 years) may result in vegetation homogenization and tree encroachment (Abreu et al., 2017; Pinheiro et al., 2016). Thus, prescribed fires may be required.

Management decisions can be improved by identifying the main vegetation and microclimate factors determining fire behaviour (Parks et al., 2018; Price et al., 2013). Both vegetation and microclimate factors are good predictors of the fire behaviour, but the relative importance between these factors can vary according to the type of vegetation and microclimatic region (Alvarado et al., 2017; Lehmann et al., 2014; McKenzie and Kennedy, 2012; Price et al., 2013). For example, the grasses abundance stands out in some savannas in arid zones, in contrast to other factors, in determining the fire behaviour (especially fire intensity) and in less arid zones the microclimate stands out in determining the fire behaviour (Lehmann et al., 2014; Lehmann et al., 2011). Similarly, in some forested environment like the Amazonia, where grasses are almost absent, microclimate is the main determinant (Balch et al., 2015; Ray et al., 2005; Uhl and Kauffman, 2012). Although it is well known that the mean woody cover, type and amount of fuel, and microclimate are important predictors of fire behaviour (Miranda et al., 2010), to the best of our knowledge, no published studies have quantified the relative importance of these factors in determining fire behaviour taking into account the range of vegetation types in the Cerrado.

In the Cerrado, experimental fires have been carried out covering different vegetation types, such as grassland, savanna and forest, and with distinct microclimatic characteristics (Gomes et al., 2018; Miranda et al., 2010). However, these studies have generally considered only one type of vegetation and one microclimate measurement (as temperature and humidity) at the time of burning (Gomes et al., 2018). Thus, the general patterns and ecological thresholds of fire behaviour, effects of microclimatic variations, and main determinants of these processes between the types of vegetation remain unclear.

Understanding the relationships between the behaviour and effects of fire in heterogeneous landscapes and with limited information, such as in the Cerrado, provides a useful basis for improving global fire risk models and our knowledge of how fire susceptibility differs around the world (Krawchuk and Moritz, 2011). Fire risk indices used in global models include the McArthur Index (FFDI) (McArthur, 1967), the Canadian Fire Weather Index (FWI) (Van Wagner, 1987), and the Keetch-Byram Drought Index (KBDI) (Keetch and Byram, 1968). These indices typically include microclimate and fuel load (Mbow et al., 2004; Perry, 1998). However, few efforts have been made to adapt these indices for the Brazilian Cerrado (Gomes et al., 2018; Nogueira et al., 2017).

In this study, we compiled literature data from 65 prescribed fire experiments in the grasslands, savannas, and forests of the Brazilian Cerrado to determine (1) how aspects of fire behaviour (fire spread rate,

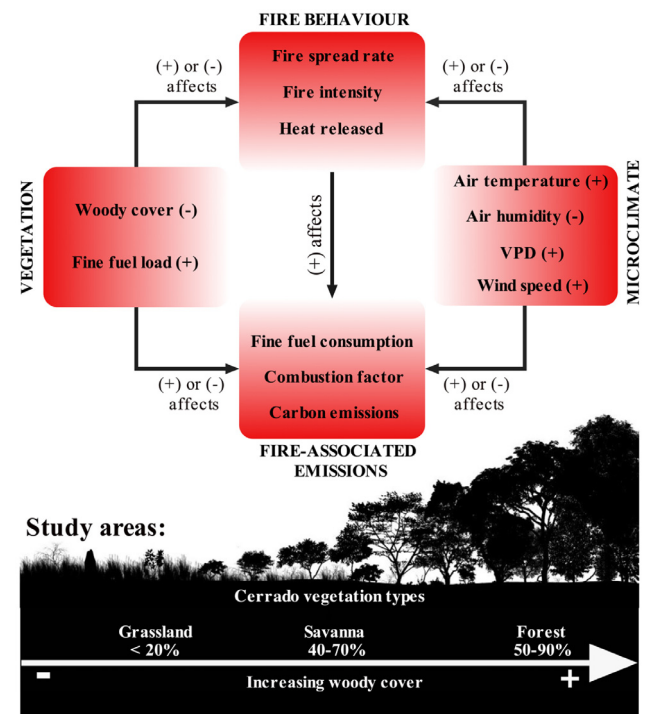


Fig. 1. Conceptual model of causal relationship between predicts and response variables (fire behaviour and fire-associated emissions) to be investigated in the Brazilian Cerrado. (+) symbol = positive relationship between boxes and (-) symbol = negative relationship. Woody cover = percentage of woody cover; fine fuel load = amount of fine fuel available for burning; air temperature = average air temperature; air humidity = relative air humidity; VPD = vapour pressure deficit; wind speed = average wind speed; fire spread rate = linear rate of fire spread; fire intensity = energy output rate per unit length of fire front; heat released = released heat per unit area; fine fuel consumption = amount of fine fuel consumed by fire; combustion factor = proportion of biomass consumed by fire; and carbon emissions = amount of released carbon associated with fine fuel consumption.

fire intensity, and heat released) and fire-associated emissions (fine fuel consumption, combustion factor, and carbon emissions associated with fine fuel consumption) vary according to the vegetation type (grassland, savanna, and forest) and (2) the relative importance of vegetation and microclimatic factors in determining the fire behaviour and fire-associated emissions. A conceptual model of the expected relationships between predictors and response variables to be evaluated in this study is presented in Fig. 1.

2. Materials and methods

2.1. Database construction

We carried out a literature review of papers published from 1994 to 2016 by searching the following databases: Web of Science, Science Direct, Google Scholar, and the Brazilian Digital Library of Theses and Dissertations. We used the following keywords in both English and Portuguese: (prescribed* burn OR fire* OR burn) AND (behavior* OR effect* OR management* OR regime* OR dynamic* OR emissions). The fire behaviour and effect variables were obtained from studies conducted in 65 plots located in six regions across the Cerrado (Fig. 2, Table S1): 1) IBGE Ecological Reserve, Distrito Federal (Castro-Neves, 2000; Castro & Kauffman, 1998; Fidelis & Pivello, 2011; Kauffman et al., 1994; Medeiros, 2002; Miranda et al. 1996; Silva, 1999; Sato, 1996, 2003), 2) Serra do Tombador Nature Reserve, Goiás (Gorgone-Barbosa et al., 2015; Rissi, 2016), 3) Chapada das Mesas National Park, Maranhão (Schmidt et al., 2016b), 4) Jalapão State Park, Tocantins

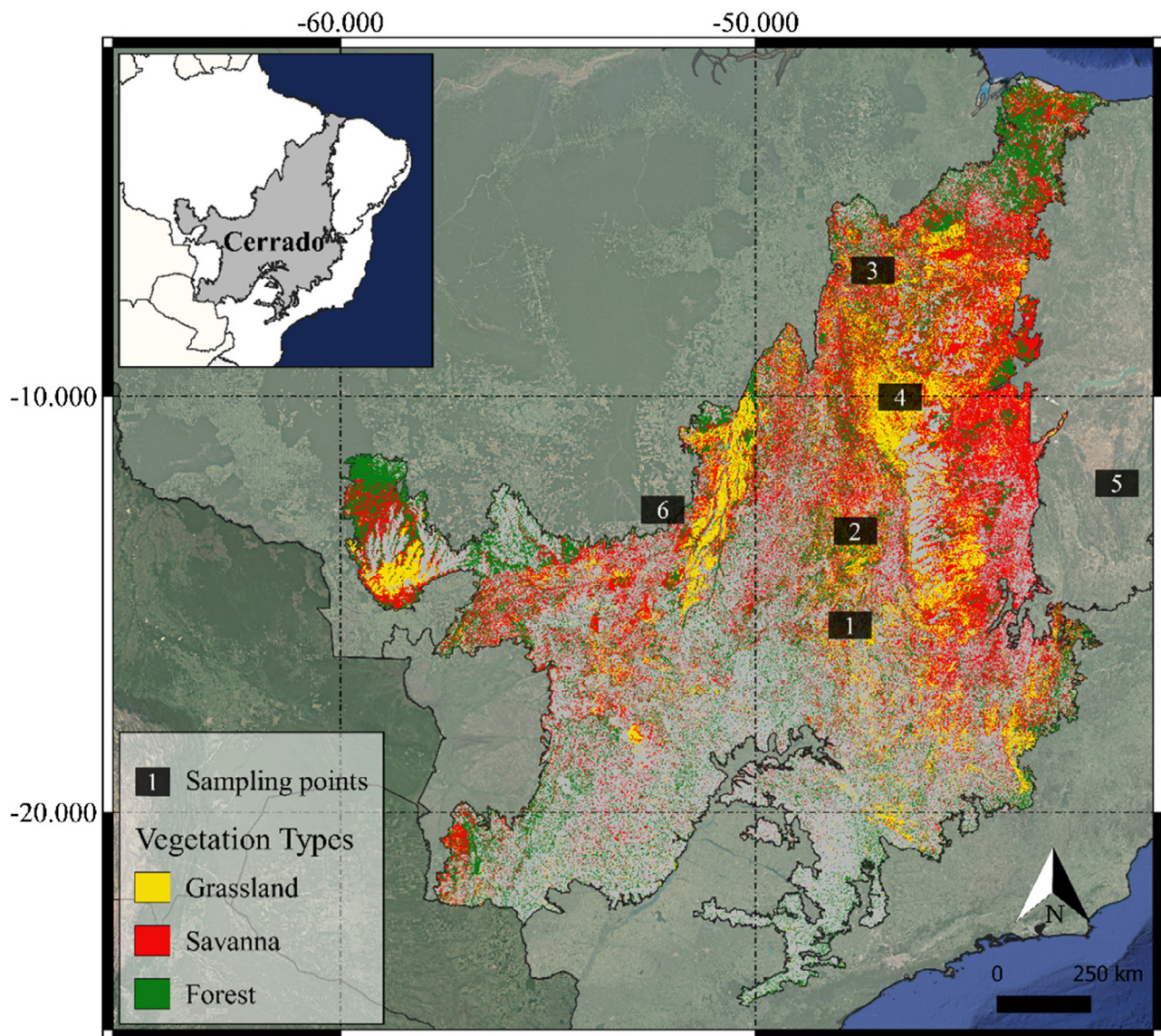


Fig. 2. Location of experimental fires carried out in grasslands, savannas, and forests of the Brazilian Cerrado from 1994 to 2016. Sampling points: 1) IBGE Ecological Reserve, Distrito Federal, 2) Serra do Tombador Nature Reserve, Goiás, 3) Chapada das Mesas National Park, Maranhão, 4) Jalapão State Park, Tocantins, 5) Serra do Sincorá, Bahia, and 6) Tanguro Ranch, Mato Grosso. The vegetation cover distribution was obtained from MapBiomas (2018) (<http://mapbiomas.org/>), and descriptions of the study areas are shown in Table S1. Colour figure only for the online version.

(Schmidt et al., 2016a, b), 5) Serra do Sincorá, Bahia (Conceição and Pivello, 2011), and 6) Tanguro Ranch, Mato Grosso (Balch et al., 2008; Brando et al., 2014). These plots were subjected to experimental fires between April and September, which corresponds to the dry season in this biome (Silva et al., 2008). Fire behaviour and its effects on the fire associated-emission were determined for the three main types of vegetation in the Cerrado biome: (1) grasslands with percentage of woody cover < 20% (36 plots), (2) savannas with percentage of woody cover ranging from 40 to 70% (19 plots), and (3) forest with percentage of woody cover ranging from 50 to 90% (10 plots) (Fig. 1). Because of the small number of experimental fire studies in Cerrado forests, we included data from a semi-deciduous seasonal forest in the transition between the Cerrado and Amazon biomes (Balch et al., 2008).

We defined fine fuel load (kg m^{-2}) as all living or dead material accumulated on the soil surface. This included grasses, leaves, and small branches ≤ 0.6 cm in diameter (Luke and McArthur, 1978) and ≤ 2 m from the ground (Castro and Kauffman, 1998), based on the low fine fuel consumption efficiency of the tree-shrub layers above that height (MCT, 2002). We did not include the following in our analysis

because the compiled studies did not contain this information: impacts on woody vegetation, topography, humidity of the fine fuel, or relationships between live and dead fuel. However, all the prescribed burned plots are on nearly flat terrain. In some of the studies, dead fuel was classified according to its time lag, which is the nominal amount of time needed to reach equilibrium moisture content in response to ambient weather conditions (Cohen and Deeming, 1985). The time lag categories used to characterise fire behaviour were 1, 10, 100, and 1000 h, which correspond to branch diameters of 0–0.64, 0.64–2.54, 2.54–7.62, and 7.62–20.32 cm, respectively. To standardise the methods and allow comparisons between samples, we selected the first diameter class (0–0.64 cm) or the time lag of 1 h and summed the amount of living or dead grasses and leaves.

In our models, we considered the following microclimate variables: 1) average air temperature ($^{\circ}\text{C}$), 2) relative air humidity (%), 3) vapour pressure deficit (VPD, kPa), and 4) wind speed (m s^{-1}). For fire behaviour, we considered: 1) fire spread rate (m s^{-1}), 2) fire line intensity ($\text{kJ m}^{-1} \text{s}^{-1}$), and 3) heat released per unit area (kJ m^{-2}). For fire-associated emissions, we considered: 1) fine fuel consumption by fire

Table 1

Fire behaviour response variables and their effects on fire-associated emissions (uncommitted emissions) between grassland, savanna, and forest of the Cerrado biome. Data are expressed as mean (standard deviation). Significant differences between vegetation types (one-way ANOVA) are shown in bold, and different superscript letters represent significant differences (Tukey's post hoc test, 95% confidence interval).

Response variable	Grassland	Savanna	Forest	df	F	P
Fire behaviour						
Fire spread rate, m s^{-1}	0.25 (0.30) ^A	0.24 (0.14) ^A	0.01 (0.00) ^B	2	65	< 0.01
Fire intensity, $\text{kJ m}^{-1} \text{s}^{-1}$	2215 (1358) ^A	2201 (3065) ^A	21 (22) ^B	2	98	< 0.01
Heat released, kJ m^{-2}	9192 (3065) ^A	9013 (3258) ^A	3028 (1591) ^B	2	18	< 0.01
Fire-associated emissions						
Fine fuel consumption, kg m^{-2}	0.59 (0.20) ^A	0.58 (0.21) ^A	0.15 (0.09) ^B	2	54	< 0.01
Combustion factor, %	87 (11) ^A	86 (10) ^A	39 (18) ^B	2	67	< 0.01
Carbon emissions, kg m^{-2}	0.23 (0.08) ^A	0.21 (0.07) ^A	0.05 (0.03) ^B	2	62	< 0.01

(kg m^{-2}), 2) combustion factor, which is the proportion of biomass consumed (%), and 3) amount of carbon emissions (kg m^{-2}) associated with fine fuel consumption (Fig. 1).

We modelled fire behaviour and fire-associated emissions using the equations shown in Table S2. The intensity of the fire front was determined using the equation proposed by Byram (1959) using an effective heat of combustion of 15,500 (kJ kg^{-1}) for grassland and savanna (Griffin et al., 1984) and 20,000 (kJ kg^{-1}) for forest formations (Pompe and Vines, 1966). The heat released was determined according to the equation proposed by Rothermel and Deeming (1980), and the amount of carbon emissions was determined according to the proportions suggested by the Intergovernmental Panel on Climate Change (MCT, 2002) based on the amount of fine fuel consumed at the time of burning (i.e. uncommitted emissions over time), fraction of oxidised biomass (live and dead) and fraction of carbon in biomass (live and dead). VPD was calculated from air temperature and relative humidity using equations from Allen et al. (1998).

2.2. Data analysis

We first explored how fire behaviour and fire effects on carbon emissions differed between the vegetation types (grassland, savanna, and forest) using one-way analysis of variance (ANOVA) (Gotelli and Ellison, 2011). We tested the homogeneity of variance (Levene's test) and normality of the distribution by evaluating residuals (Shapiro-Wilk's test) using the "aov" function of the "stats" package in R software (R Core Team, 2019).

To evaluate the relative importance of vegetation and microclimatic factors in determining the fire behaviour and fire-associated emissions, we first remove the non-independent or collinear variables. We tested for multicollinearity between explanatory variables using the "vif" function of the "car" package in R. The explanatory variables with high collinearity (variance inflation factor [VIF] greater than 10) were removed from the analyses (Draper and Smith, 1998). The predictors of fire behaviour assessed by the VIF were vegetation type (corresponding of woody cover), air temperature, relative air humidity, VPD, wind speed, and fine fuel load (Fig. 1). As expected, air temperature and relative air humidity were highly collinear with VPD. The predictors of fire-associated emissions assessed by the VIF were vegetation type, fine fuel load, air temperature, relative air humidity, VPD, wind speed, fire spread rate, fire intensity, and heat released (Fig. 1). Whereby, heat released and fire intensity were collinear with fine fuel load. Thus, for fire behaviour models, we considered only vegetation type, VPD, wind speed, and fine fuel load. For carbon emission models, we considered only vegetation type, fine fuel load, VPD, wind speed, and fire spread rate.

Following this selection, we constructed generalised linear models to evaluate fire behaviour and fire-associated emissions as functions of vegetation and microclimate factors. We also tested potential interactions between vegetation type and each of the other variables. For each

vegetation type, we elaborated a model for each fire behaviour characteristic and for the fire-associated emissions variable.

We used the Akaike information criterion corrected for small samples (AICc) to select the models, with $\Delta\text{AICc} < 2$ indicating little to no difference between models. We identified the best model on the basis of Akaike weight (wAICc), which represents the probability that a set of variables results in the best fit using " dredge" function of the "MuMIn" package in R (Burnham and Anderson, 2002). For the final model, we verified the assumptions of the analysis by testing the homogeneity of variance (Levene's test) and normal distribution of the residuals (visual assessment) (Burnham and Anderson, 2002). We determined the relative importance of each predictor variable selected for the global model from the sum of Akaike weights (w) using the "importance" function of the "glmulti" package in R. We also calculate the linear relationships between these same predictors using "rsq" and "rsq.partial" functions of the "rsq" package in R.

3. Results

Fire behaviour response variables and fire-associated emissions differed between vegetation types ($P < 0.05$; Table 1). The fire spread rate was higher in grassland and savanna than in forest (0.25 vs. 0.01 m s^{-1}). The same pattern was observed for fire intensity (2208 vs. 21 $\text{kJ m}^{-1} \text{s}^{-1}$), heat released (9103 vs. 3028 kJ m^{-2}), fine fuel consumption (0.59 vs. 0.15 kg m^{-2}), combustion factor (86% vs. 39%), and carbon emissions associated with fine fuel consumption (0.22 vs. 0.05 kg m^{-2}).

Fire behaviour and fire-associated emission varied according to vegetation type (grassland, savanna, and forest) and microclimate. Regarding fire behaviour, fire spread rate was negatively affected with relative air humidity (for all vegetation types) and positively correlated with air temperature (for savanna and forest), VPD (for grassland, savanna, and forest), and wind speed (for grassland) (Fig. S1). Fire intensity was negatively correlated with air humidity (for grassland and savanna) and positively correlated with VPD (for grassland and savanna), wind speed (for grassland and savanna), and fine fuel load (for savanna and forest) (Fig. S2). The heat released was positively correlated with fine fuel load (for grassland and savanna) (Fig. S3). Regarding fire-associated emission, fine fuel consumption was negatively correlated with relative air humidity (for savanna) and positively correlated with fine fuel load (for grassland and savanna), fire intensity (for savanna and forest), and heat released (for all vegetation types) (Fig. S4). The combustion factor was positively correlated with fire intensity (for savanna and forest) and heat released (for all vegetation types) (Fig. S5). Finally, the amount of carbon emission was negatively correlated with relative air humidity (for grassland and savanna) and positively correlated with fine fuel load (for grassland and savanna), fire intensity (for savanna and forest), and heat released (for all vegetation types) (Fig. S6).

In general, both fire behaviour and fire-associated emissions were

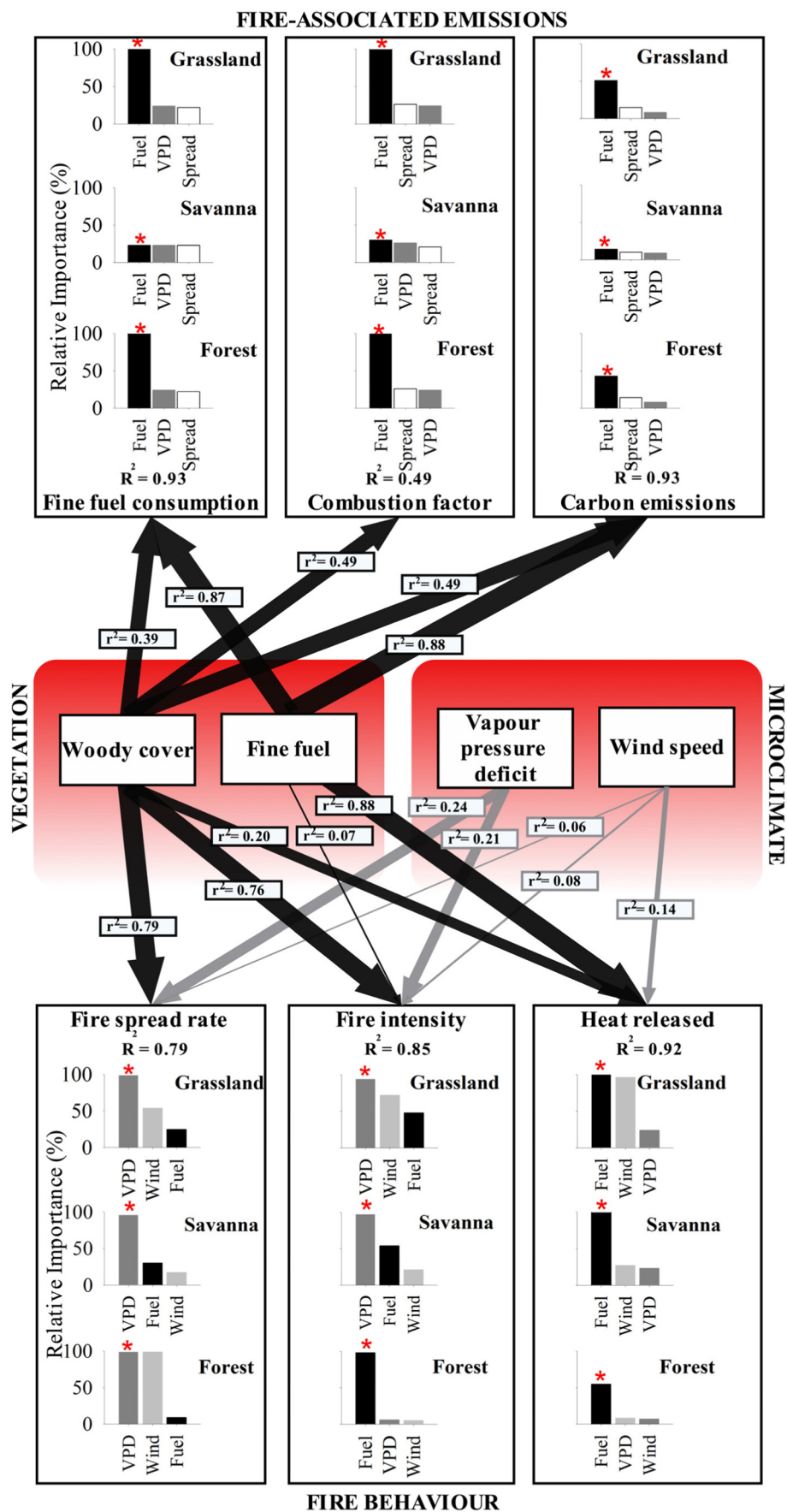


Fig. 3. Models for fire behaviour and fire-associated emissions (uncommitted emissions) as a function of vegetation type and microclimate in grassland, savanna, and forest of the Brazilian Cerrado. Arrows represent significant linear relationships ($P < 0.05$). R^2 = coefficient of determination, r^2 = partial coefficient of determination, *primary driver. Fuel = fine fuel load, VPD = vapour pressure deficit, Wind = wind speed.

explained primarily by the same predictors in open formations (savanna and grassland) and forest (Fig. 3). Regarding fire behaviour, fire spread rate was explained primarily by VPD in all vegetation types. Fire intensity was explained primarily by VPD in open formations and by fine fuel load in forest. The heat released was explained primarily by fine fuel load in all vegetation types. Most of the best models for fire behaviour included VPD as a predictor (Table S3), whereas most of the best models for fire-associated emissions included fine fuel load as the only predictor (Table S4). The relative importance of vegetation and microclimate factors as predictors varied more with respect to the type of response variable evaluated (fire behaviour or fire-associated emissions) than with vegetation type.

The variables that contributed the most to the models were vegetation type, VPD, and wind speed (for fire spread); vegetation type, fine fuel load, VPD and wind speed (for fire intensity); vegetation type, fine fuel load, and wind speed (for heat released); vegetation type and fine fuel load (for both fine fuel consumption and carbon emission); and vegetation type (for combustion factor) (Fig. 3, Table S5). The equations generated by model selection are shown in Table S6.

4. Discussion

Our results showing higher fire spread rate, fire intensity, heat released, fine fuel consumption, combustion factor, and carbon emissions associated with fine fuel consumption in grasslands and savannas compared to forests can be explained by differences in vegetation structure and microclimate. The open formations (grasslands and savannas) are characterised by sparse vegetation (Ribeiro and Walter, 2008), higher wind speed, higher air temperature, and lower air humidity, which favour fire propagation (Hoffmann et al., 2012). In addition, the fine fuel in open formations consists of a continuous layer of grasses (Hoffmann et al., 2012; Hoffmann and Franco, 2003), which are highly flammable (Cardoso et al., 2018; Wragg et al., 2018). In contrast, grasses rarely occur in forests, and the fine fuel moisture content tends to be higher (Charles-Dominique et al., 2018; Hoffmann et al., 2012), which reduces the fire spread rate and fine fuel consumption (Ray et al., 2005). However, the proportion of grasses and fuel continuity were not measured in most of the studies compiled here. These results indicate that fire behaviour is better explained when we evaluate microclimate and fine fuel together with vegetation structure.

There was little difference in the relative importance of vegetation and microclimate factors in determining fire behaviour and fire-associated emissions among vegetation types. We had expected that fine fuel load would be the most important factor in grassland and savanna, and VPD would be the most important factor in forest. However, variations in fine fuel load had little effect on the fire spread rate or fire intensity in the grassland and savanna, whereas higher VPD was correlated with higher fire spread rate and fire intensity in these vegetation types. We found that fine fuel load was less important in determining fire spread rate when comparing environments with a similar quality of flammable material (e.g. proportion of grasses).

The relative importance of vegetation and microclimate variables did vary according to type of response variable (fire behaviour or fire-associated emissions), which had not previously been reported in the Cerrado. Thus, fire management must take into account multiple predictors and response variables. For example, a lower fire spread rate may result in greater reduction in tree and shrub biomass, because a slower fire can ignite the bark, seriously damaging the vascular cambium tissue and resulting in mortality (Kayll, 1968; Michaletz and Johnson, 2007; Silva and Miranda, 1996). In contrast, a higher fire intensity and heat released may result in greater reductions in ground layer biomass (Trollope et al., 2002). Furthermore, consumption of flammable material, combustion factor, and carbon emissions are the primary metrics for evaluating fire-related pollution and greenhouse gas emissions (MCT, 2002).

Although VPD has been little used in global fire models (Drüke

et al., 2019), our data showed that fire spread and fire intensity were primarily related to VPD, with an additive effect of wind speed, which contribute to fire behaviour through fuel drying and fuel pre-heating, respectively (Cheney and Sullivan, 2008). However, VPD was not a good predictor of heat released, fine fuel consumption, combustion factor, or carbon emission, which were primarily explained by fine fuel load. These results highlight the importance of considering multiple vegetation and microclimatic factors in fire management strategies (Hoffmann et al., 2012).

Our study provides support for decision making related to fire management and biodiversity conservation policies in the Cerrado. For open formations, we recommend monitoring VPD and wind speed to plan prescribed fires with a higher fire spread rate and smaller impacts on woody vegetation. Burns with rates of spread greater than 0.4 m s^{-1} would require a VPD of about 3 kPa, which corresponds approximately to an air temperature of 30°C , relative humidity of 30%, and wind speed greater than 2 m s^{-1} . For fires at the beginning of the dry season, when the VPD is lower, a higher wind speed is required ($\geq 3 \text{ m s}^{-1}$). However, rapid burns require greater control efforts to prevent spreading beyond the planned area. For open formations we also recommend conducting mosaic burning with different frequencies and periods of burning to allow the establishment of endemic species unique to each fire regime (Pivello, 2011). The construction of firebreaks can reduce the extent of the burned area and consequently reduce total carbon emissions. For forests we recommend monitoring the fine fuel load, VPD, and wind speed to determine the risk of forest fires.

5. Conclusion

Fire spread rate, fire intensity, heat released, fine fuel consumption, combustion factor, and carbon emissions from fine fuel consumption are higher in grasslands and savannas than in forests. For all vegetation types, VPD was the main predictor of fire behaviour, whereas fine fuel load was the main predictor of fire-associated emissions. Thus, management strategies in Cerrado should take into account the effects of vegetation and microclimate on fire behaviour and fire-associated emissions. Changes in VPD associated with climate change may significantly impact fire behaviour in all vegetation types in the Cerrado and must be considered in current and future strategies for prescribed burns in the region.

CRedit authorship contribution statement

Letícia Gomes: Conceptualization, Methodology, Data curation, Formal analysis, Visualization. **Heloisa S. Miranda:** Conceptualization, Methodology, Visualization, Investigation. **Divino V. Silvério:** Visualization, Investigation, Formal analysis. **Mercedes M.C. Bustamante:** Conceptualization, Methodology, Visualization, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2019.117804>.

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